Shakib Kafashan

Transportation, Mobility & Simulation

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Summary

Transportation Systems Ph.D. candidate at UC Irvine studying autonomous-fleet routing, traffic equilibrium, and probabilistic vehicle-interaction safety. Combines network optimization, Monte Carlo simulation, naturalistic trajectory evaluation, and statistical analysis to assess mobility and safety under uncertainty.

Technical Skills

Programming & spatial analysis: Python; ArcGIS; QGIS

Transportation modeling: Network routing; Wardrop user equilibrium; bilevel network design; mode-choice modeling; vehicle-interaction geometry

Simulation & uncertainty: Monte Carlo simulation; motion uncertainty; sensitivity analysis; distributionally robust receding-horizon control

Safety analytics: Naturalistic trajectory evaluation; logistic regression; agency-clustered inference; temporal stability testing

Selected Research Experience

University of California, Irvine | Institute of Transportation Studies

Safety-Aware Autonomous-Fleet Routing

Ongoing research | INFORMS Annual Meeting 2026 presentation

- Developing a safety-aware network-routing framework for autonomous fleets that accounts for the equilibrium response of human-driven traffic.
- Preparing an INFORMS 2026 presentation on the research, with a focus on connecting fleet-routing decisions, traveler response, and network safety.

Vehicle-Tube Safety Assessment Under Motion Uncertainty

First-author research | Submitted to Accident Analysis & Prevention

- Evaluated a geometric model of vehicle-body contact under uncertain motion, using prescribed paths and reaction assumptions for following, crossing/turning, and merging interactions.
- Assessed 1,118 naturalistic interactions: 777 highD following, 112 inD crossing/turning, and 229 exiD merging cases.
- Used Monte Carlo simulation, numerical refinement, and sensitivity studies to characterize variation in the contact-risk score across interaction types.
- Benchmarked model scores against observed minimum body clearance and standard proximity metrics to characterize performance across interaction types.

Equilibrium-Based Detour-Sign Placement

Coauthor and corresponding author | Under review, Transportation Research Record

- Contributed to a bilevel network-design study minimizing vehicle-miles traveled on vulnerable links while modeling traveler response through user equilibrium.
- Study benchmarks achieved relative optimality gaps below 1% on four networks with 76-914 links, using screened candidate sets of 10-20 links.

Selected Research Experience (continued)

Autonomous Freight Operations & Fallback Reliability

First-author research | Full draft in preparation

- Developed a receding-horizon freight controller that coordinates uncertain drone capacity with a finite allowance of hazardous human exposure in a two-node Everest corridor.
- Combined empirical weather inputs with synthetic shipment calendars and designed disruption scenarios; compared six operating policies in 540 policy-season simulations.
- Isolated the reserve-policy effect in 450 matched transition-regime seasons: mean critical-service gain was 0.704 percentage points, rising to 1.625 points under high fallback pressure.
- Verified preserve-now/use-later behavior in 71 seasons, typically shifting 21 person-hours from routine to later critical freight use.

Transit-Bus Safety & Vulnerable Road Users

First-author research | Manuscript in preparation

- Constructed a casualty-level analytical cohort from National Transit Database event records and linked operating data: 5,915 casualties, 5,786 events, and 305 agency IDs, 2014-2025.
- Estimated binary logit models with agency-clustered inference and tested temporal stability to distinguish conditional fatal-injury severity from collision-occurrence risk.
- Quantified an adjusted senior-versus-adult fatal-severity difference of 10.82 percentage points, with agency-clustered uncertainty and checks for reporting and temporal differences.

Walkability & Travel Mode Choice

Sharif University of Technology | 2021-2022

- Engineered GIS walkability features, combined them with trip records, and analyzed mode choice using logit and machine-learning models.
- Used Python and ArcGIS/QGIS to connect built-environment measures with observed travel behavior.

Education

University of California, Irvine

Ph.D., Civil Engineering (Transportation Systems) — Sep 2024 – Present

M.S., Civil Engineering (Transportation Systems) — Sep 2022 – Jun 2024

Sharif University of Technology

B.Sc., Civil Engineering — Jul 2018 – Jul 2022

Selected Publication

Kafashan, S.; Saphores, J.-D. (2026). Building Sustainably: Annualized Cost of Ownership, Externalities, and the Electrification of Construction Machinery. *Sustainability*, 18(12), 6343. doi:10.3390/su18126343.